R13

Code No: 126AQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
INFORMATION SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper Aug-2024

PART - A

(25 Marks)

1.a)	Define Computer Security and list the three factors of Computer security.	[2]
b)	List the characteristics of Cryptographic techniques.	[3]
c)	Define Substitution and Permutation.	[2]
d)	Differentiate between Block Cipher and Stream Cipher.	[3]
e)	Define hash function and list the functionalities of hash function.	[2]
f)	Write short notes on Biometric Authentication.	[3]
g)	What is the need of Email Security?	[2]
h)	Mention various modes of Authentication header and explain them.	[3]
i)	Define Secure Socket Layer.	[2]
j)	List any four firewall design principles.	[3]

PART - B

(50 Marks)

2.a) Mention and explain various types of Security Services.
b) With an example, explain any two types of Transposition techniques. [5+5]

OR

3. Define Threat and attack. Differentiate between them. List some examples of attacks which raise in real world cases and explain. [10]

4.a) Draw and explain the single round of DES algorithm.
b) Write and explain the principles of public key cryptography. [5+5]

OR

5. Why mode of operation is defined? Explain the Block Cipher modes of operation. [10]

6.a) Is Message Authentication code same as encryption? How message authentication can be done by message Authentication code? Explain with an example.
b) With a neat block diagram, explain Digital Signature algorithm. [5+5]

OR

7. What is key wrapping and key Unwrapping? Write and explain key wrapping algorithm. [10]

8.a) List and explain various Services provided by Email Security.
b) How to encrypt Email with S/MIME? Explain. [5+5]

OR

9. With a neat sketch, explain the architecture of IP Security. [10]

10. Which Parameters define Session state and which parameters define connection state in SSL? Explain. [10]

OR

11. How can we achieve web security? Explain Web Security threats with an example. [10]

---ooOoo---

used paper Aug-2024